Far-Field *Singing* FPGAs: Repurposing Routing Fabrics into 100 m Covert Radiators

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***Abstract*—FPGAs rely on highly dense and symmetric internal routing networks to interconnect their configurable logic elements. In standard applications, these interconnects are used solely for digital signal transfer within the device, leaving many routing paths idle. We study the surprising ability of configurable FPGA routing fabrics to act as intentional radiators when structured and driven coherently. Building on prior near-field demonstrations (few centimeters), we (i) present a practical toolchain and methodology for synthesizing "fabric-only" antennas using constrained placement/routing; (ii) demonstrate reliable far-field reception for extremely long ranges ($\leq$ 100 m) and quantified bit-error performance at meter-scale ranges using ASK/FSK modulation and simple ECC; and (iii) analyze the security implications by formalizing adversary capabilities, enumerating novel multi-tenant attack vectors, and outlining detection and mitigation strategies. Our work bridges implementation engineering, complex physical-layer measurement (with a set of complex Far-Field measurement apparatus), and security analysis, and highlights the urgent need for screening and runtime monitoring in shared FPGA environments. We have systematically shaped and combined *unused* paths into a contiguous structure, such as {Fractal, loop, Dipole, Snake, Spiral, Array}-Antennas, which required building an automation tool-chain. When energized, this embedded structure emits measurable electromagnetic energy that can serve as a stealth communication channel. We've extended this concept far beyond previous near-field demonstrations, achieving reliable reception in the Far-Field, demonstrated rigorously with various measurements setups - a first for this class of long-range FPGA-based antennas without any external radiating RF hardware from a tiny $\sim$ 1x1 cm^2 device. We further show a Trojan example while triggering it with rare events attacking a Decryption Oracle model.**

***Index Terms*—FPGA interconnect, embedded antenna, covert antenna, electromagnetic emission, far-field transmission, hardware security, side-channel communication, routing resources, unconventional antennas.**

I. Introduction

Field-Programmable Gate Arrays (FPGAs) are highly versatile integrated circuits composed of an array of reconfigurable logic blocks, surrounded by a rich network of programmable interconnects and configurable input/output interfaces. Unlike fixed-function chips, their functionality is defined after manufacturing by loading a configuration file, allowing designers to implement a wide variety of digital circuits on the same physical device. This reconfigurability, combined with fine-grained parallelism, enables FPGAs to deliver high performance [1], [2] for specialized tasks while retaining flexibility for updates or design changes. As a result, they have gained widespread popularity across domains such as signal processing, telecommunications, artificial intelligence acceleration, embedded systems, prototyping of ASIC designs, and even high-performance cloud computing. Their ability to bridge the gap between software adaptability and hardware efficiency makes them a critical technology in both research and industry. Concretely, FPGAs are widely deployed today across edge, enterprise, and cloud infrastructures for their programmability and performance which serves the main threat and motivation for the outlined research, *they may expose physical resources that can be repurposed in unexpected ways*.

A limited research effort has explored the creation and analysis of **unconventional or covert antennas** within FPGA fabrics, primarily utilizing underutilized programmable interconnects to form radiative structures not intended in conventional digital design [3]–[5]. These demonstrations initially focused on near-field effects; in this paper we extend those observations to show that with careful design and tooling, the FPGA fabric alone (no package pins or external RF components) can radiate sufficiently to allow far-field reception with commodity antennas and SDRs. It highlights possible multi-tenant and cloud FPGA threats as well as other possible interplay with other recent RF threats [6], [7].

Contributions: Our work introduces several key advancements that differentiate it from prior research on FPGA-based antennas. Most notably, we achieved successful far-field reception of signals radiated from FPGA internal wiring at distances exceeding 100 meters-significantly surpassing the typical near-field ranges reported in earlier studies as listed in Table I. This milestone demonstrates the practical feasibility of using FPGA internal interconnects not just as exotic antennas, but as effective transmitters capable of long-distance wireless communication. We did that by implementing a complete transmission pipeline (PLL-derived excitation, dividers, modulators, and repetition-based ECC) and experimentally characterized five antenna geometries across near- and far-field measurement setups, presenting SNR and BER curves and a simple discrete-channel model as listed in Table II. Additionally, we developed an intuitive and user-friendly EDA tool that allows straightforward creation of internal FPGA antenna geometries by visually drawing shapes mapped to routing resources, translating user-drawn geometries into synthesizable Verilog with placement and routing constraints, dramatically lowering the barrier to fabric-only antenna design as listed in Table III. We also implemented frequency shift keying (FSK) and amplitude shift keying (ASK) modulation on these embedded antennas, showcasing their capability to carry robust digital data.

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Together, these contributions open new avenues for both tool-assisted FPGA antenna design and expanded wireless communication applications, pushing the boundaries of what internal FPGA structures can accomplish. Given these results, we formalize the adversary and threat models relevant to multi-tenant and on-premise FPGAs and enumerate realistic and novel attack vectors - including collaborative multi-tenant schemes, phased-array-style collusion across multiple board instances, and stealthy harmonic-hopping strategies intended to evade spectral monitors.

The remainder of this paper is organized as follows. Section II places our work within prior art. Section III describes the design methodology and EDA tooling as well as details on the experimental setup and measurement procedures. Section IV presents and analyzes the results. Section V discusses reception filtering and simple error correction errorrates. Section VII formalizes the adversary model and Section VIII enumerates plausible attack vectors and mitigation strategies. We conclude in Section X with a discussion and future directions.

II. Related Work

Unconventional antennas synthesized from digital wiring and logic have been reported in multiple studies [3]–[5]. These works established that net geometry, repeaters/buffers, and placement influence emission patterns and near-field strength. Our work extends these findings by demonstrating far-field reception from fabric-only radiators, by automating geometry-to-Verilog mapping, and by providing an extensive security analysis tailored to multi-tenant deployments. We also draw on antenna theory and EM measurement methodologies to quantify SNR and far-field behavior in sub-resonant, electrically short structures.

Tavaragiri *et al.* and Couch *et al.*, described methods to concatenate unused wire segments into pseudo-equipotential geometric shapes, such as rectangles and fractals [3], [4]. These net structures, generated via custom toolchains (e.g., TORC toolkit and fill routers), could be excited with oscillating signals to produce detectable electromagnetic emissions. Systematic experimentation demonstrated that both the **shape** and **size**, quantified by the number of wire segments, impacted the frequency response and radiative efficiency of the embedded antennas, with larger and longer shapes generally increasing emission magnitude [3]. Additionally, the position of the antenna within the chip and in the context of coexisting digital logic was shown to influence stability and performance (recall their near-field focus which is ultra location-sensitive).

These early works leveraged on-chip oscillators and modulation techniques such as Amplitude Shift Keying (ASK) or Binary Phase Shift Keying (BPSK) to drive these embedded antenna nets and facilitate signal detection in the near-field or quite limited analysis of greater distances using spectrum analyzers and wide-band antennas. The experimental setups typically include careful automation of power, thermal, and emission measurements, providing empirical insight into the thermal and electrical stability boundaries encountered during high-power tests [3]. The studies also highlight significant deviations from traditional antenna theory, noting that the effective lengths and re-buffering structures of FPGA wiring preclude simple analogies with classic antennas.

Recent work by Aknesil *et al.* examines the security ramifications of such techniques in the context of **covert communication** and **hardware Trojans** [5]. They extend the concept to a wider array of antenna implementations, including those made solely from FPGA interconnect, large arrays of flip-flops, and even package pins: While Aknesil et al. (2023) [5] achieved data transmissions through near-field embedded antenna networks inside the FPGA, their approach remains limited in both scope and practical relevance. The use of arrays of flip-flops and package pins relies on more identifiable or externally accessible wiring, which reduces the subtlety and threat potential of the attack. Moreover, near-field coupling inherently restricts range and scalability, making their findings less impactful for understanding realistic information leakage risks. In contrast, our work demonstrates far-field transmission capabilities that overcome these limitations, highlighting genuine long-range covert channels that originate solely within the FPGA fabric.

TABLE I
Key Works on Unconventional FPGA Antennas (Part 1)

Reference	Antenna Method	Transmission Range	Wiring Location
Tavaragiri *et al.* [4]	Interconnect geometries, fractals	Near-field ($\sim$2 cm)	Internal FPGA interconnect
Couch *et al.* [3]	Automated fill routers, multiple shapes	Near-field with wide-band antenna	Internal FPGA interconnect
Aknesil *et al.* [5]	Routing nets, flip-flop arrays, package pins	Near-field (few cm)	Interconnect + flip-flops + **package pins**
Our Work	**Routing fabric only (no external components)**	**Far-field (up to 100 m)**	**Exclusively internal routing fabric**

TABLE II
Key Works on Unconventional FPGA Antennas (Part 2)

Reference	Detection Method	Key Findings
Tavaragiri *et al.* [4]	Small loop antenna	Antenna shape impacts emission; overlays on active logic possible
Couch *et al.* [3]	Wide-band antenna	Emission and thermal baseline; practical embedded antenna toolchain
Aknesil *et al.* [5]	Near-field probe	High-accuracy covert comm.; resource and performance trade-offs
Our Work	**Multiple: near-field probe, dish, log-periodic (far-field)**	**First far-field demo (100 m); robust, fabric-only; covers realistic cloud FPGA threats**

Together, these studies establish both the enabling mechanisms and the profound **security implications** of programmable interconnects as unconventional antennas in modern FPGAs, underscoring the need for detection and mitigation in secure design methodologies [3]–[5].

III. Design Methodology and Toolchain

This section outlines the methodology and tools used to generate and preserve custom routing geometries in the FPGA fabric. It introduces the Draw-to-Route EDA tool, details the placement and routing constraints applied to maintain geometric fidelity, and describes the RTL architecture used for excitation and modulation.

TABLE III
SPECIAL PARAMETERS AND ROUTING SCHEME OF KEY WORKS ON UNCONVENTIONAL FPGA ANTENNAS

Reference	Special Parameters	Routing Technology
Tavaragiri *et al.* [4]	Use of diverse antenna geometries	Routing created by mapping a binary bitmap to FPGA tiles representing routing resources. Utilized ADB tool to identify conflicts of wires and remove them.
Couch *et al.* [3]	Automated fill routing to generate multiple antenna shapes	TORC toolkit for FPGA routing guided by a binary BMP image. The approach is FPGA-specific, does not enable parallel routing, and lacks synchronization across multiple routes.
Aknesil *et al.* [5]	Near-field routing nets, flip-flop arrays, package pins	Manual VHDL code
Our Work	**Automated fill routing of diverse antenna geometries, with parallel routing and synchronized buffering for far-field fabric-only design**	**Draw-to-Route EDA leverages XDLRC parsing and creates a GUI to automate antenna routing on any FPGA. It ensures repeater insertion, multi-LUT support and synthesizable constraints.**

A. Draw-to-Route EDA Tool

Manual creation of complex geometric structures on FPGAs requires connecting large numbers of individually specified wires and LUTs to realize a desired antenna-like pattern. This process is highly labor-intensive and error-prone. To address these challenges, a dedicated Electronic Design Automation (EDA) tool, Draw-to-Route, was developed to automate the routing specification process and generate deterministic placement and connectivity.

The tool parses the FPGA's XDLRC (Xilinx Device Resource Configuration) file, generated through Xilinx ISE or Vivado, and presents a grid-based graphical interface that reflects the physical CLB layout of the device. Within this GUI, users can "paint" the desired geometry directly onto the FPGA grid (Fig. 1). Additional controls allow users to specify:

- Repeater insertion
- Use of multiple LUTs per CLB (up to four) to increase emission strength by using more routing resources

Once drawn, the design can be exported as synthesizable Verilog with deterministic placement and routing attributes. The generated source includes directives such as `LOC` and `DONT_TOUCH` to ensure preservation of geometric intent through synthesis. The internal generator accounts for routing resources and inserts repeaters where required to maintain signal integrity across long interconnects.

Quality-of-life features such as undo, clear sketch, zoom, reset view, and "Load Verilog" facilitate iterative refinement. A "New Segment" function allows initialization of secondary or independent net paths within the same design session, enabling multi-source or multi-geometry routing experiments.

B. Forcing Deterministic Routing

Ensuring that CAD-defined geometries survive synthesis and place-and-route stages requires explicit constraints within the FPGA toolchain. Each LUT instance is annotated with Verilog attributes of the form:

```
(* S="TRUE", dont_touch="TRUE",
LOC="SLICE_XxYy" *).
```

These constraints fix every LUT and routing element to predefined spatial coordinates, forcing the physical layout to match the designed topology. In cases requiring coarse region control, floorplanning primitives were applied to pin CLB locations.

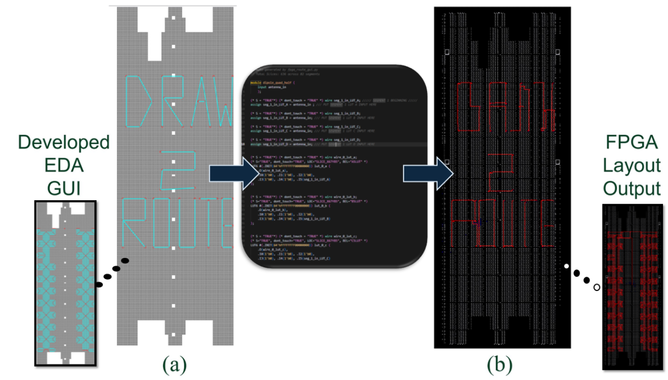

Fig. 1. Draw to Route EDA software to draw antenna's designs on FPGA

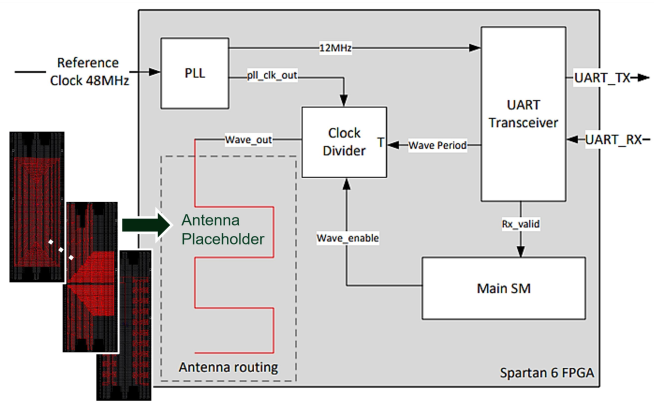

Fig. 2. Block Diagram of the FPGA Square Wave Transmission and Control Architecture

C. RTL Transmission Stack

Our RTL implements: a PLL-based high-frequency source, a programmable divider, a small control state machine, an ASK/FSK modulator, and a payload buffer with repetition coding. Fig. 2 depicts the overall architecture implemented on a Xilinx Spartan-6 FPGA to enable controlled electromagnetic emission via an internal routing structure configured as an embedded antenna. The system is driven by a 48MHz reference clock, which is supplied to a phase-locked loop (PLL) module within the FPGA. The PLL generates two output clocks: The system clock (12MHz) and the main antenna clock *pll_clk_out* (936MHz). The system clock is used by the Main SM and UART Blocks and the antenna clock drives the clock divider module, which is responsible for producing the signal that drives the antenna. The divider receives user-configurable parameter *wave_period* via UART, with input data handled by a main state machine. Upon validity of the input configuration (as indicated by the *RX_VALID* signal), the system configures the clock divider and enables the square wave output, which is routed through an antenna structure realized entirely in the FPGA's programmable interconnect. The resulting *Wave_out* signal thus traverses the selected routing fabric, facilitating

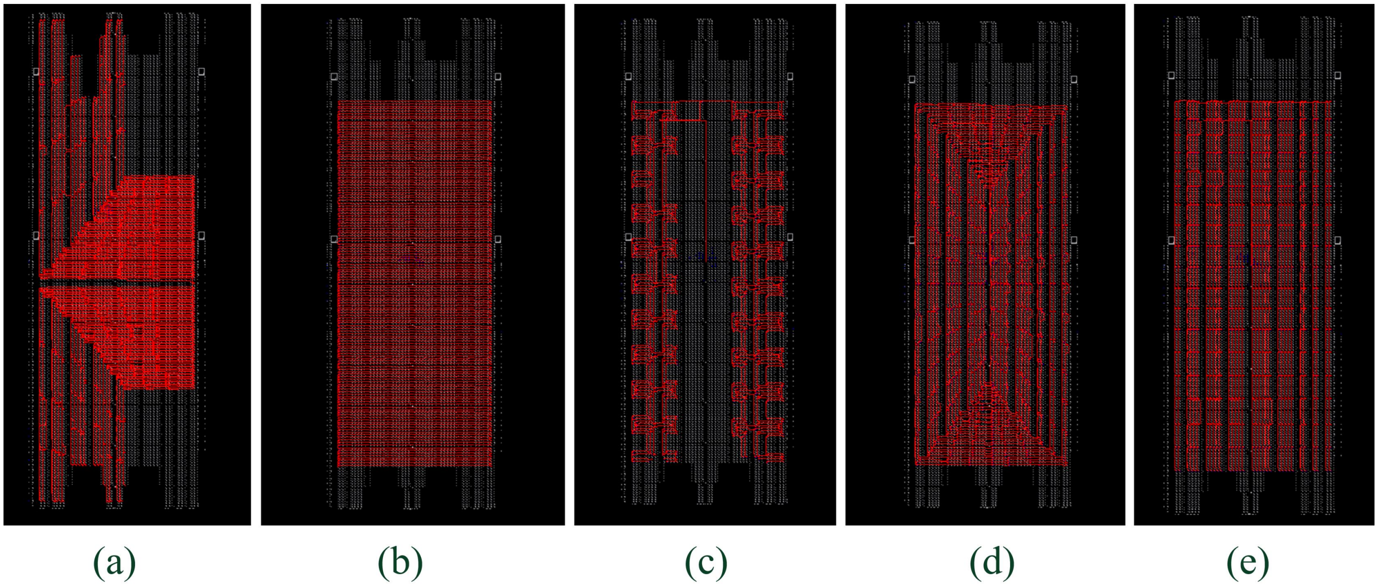

Fig. 3. (a) Dipole Antenna (b) Snake Antenna (c) Fractal Antenna (d) Spiral Antenna (e) Antenna Array as seen in the "FPGA Editor" program after place & route

electromagnetic emission that can be measured externally. The modular structure and programmability of each block enables fine-grained control over waveform frequency, essential for characterizing and optimizing the performance of in-fabric antennas. The UART interface was employed to connect the FPGA to a host Python application, enabling direct control of the on-chip state machine from a personal computer without requiring any modification to the FPGA's programmed logic.

D. Transmitter Antenna Designs

In designing the on-chip antennas, three principal constraints were taken into account. First, the total physical dimensions of the FPGA die are maximum 1×1 cm^2, with the effective area available for implementing custom LUT-based structures constrained to even less than that. This makes our antenna extremely short, and that makes the effective resonance frequency for the antenna very high (over 5GHz). Second, the detectability of radiated signals over far field is inherently limited by the reduction in field strength with distance, compounded by the finite sensitivity of the measurement equipment. Third, the maximum frequency of the driving square wave was restricted to 936MHz, based on the operational limits specified in the PLL configuration generator software - making us work much below the resonance frequency of any achievable antenna.

Using the capabilities of the our EDA tool, we designed and implemented five distinct FPGA-based antenna configurations. Each design was informed by recent advances reported in the literature and optimized with the objective of maximizing the signal-to-noise ratio (SNR) in far-field measurements. The antenna implemented designs were Dipole Antenna [8], Snake Antenna [9], Fractal Antenna [10], Spiral Antenna [11], Antenna Array [12] as illustrated in Fig. 3.

These antennas emerged as the most promising candidates based on the previously outlined design constraints. Furthermore, their implementation requires only resources commonly available on nearly all FPGA boards, rendering the designs broadly applicable and not limited to specific devices. Consequently, we selected these five antennas for empirical evaluation to determine which configuration delivers the optimal performance.

E. Receiver Antenna Designs

To evaluate the strength of the transmitted signal, our objective was to search for the highest Signal-to-Noise Ratio (SNR). This work presents our experiment's results obtained using three distinct receiving antenna types as illustrated in Fig. 4: a small magnetic Loop Antenna (Rhodes & Schwarz RSH50-1), Log-periodic Antenna (HyperLOG 30100X) and Dish Antenna (TP-Link TL-ANT2424MD).

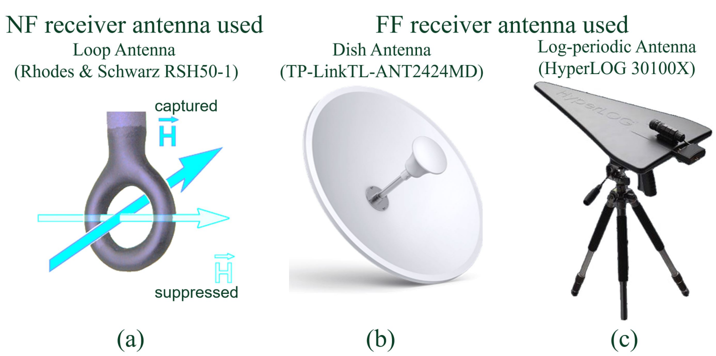

Fig. 4. Receiver antennas used in the different setups throughot the research: (a) Small magnetic Loop Antenna (b) Dish Antenna (c) Log-periodic Antenna

IV. Results and Analysis

The experimental validation comprises three sequential measurement campaigns spanning both near-field and far-field

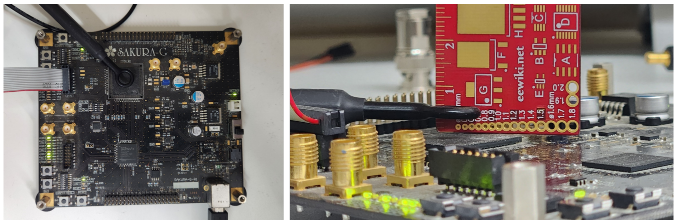

Fig. 5. Experiment A setup - Near Field small magnetic loop antenna: (**left**) top-view (**right**) side-view

regimes: (A) near-field frequency-sweep analysis to determine SNR peaks across antenna geometries, (B) far-field range analysis utilizing a dish antenna to establish maximum detection distance and asymptotic SNR behavior, and (C) far-field multi-band characterization employing a log-periodic antenna to validate signal robustness across extended separations and frequency coverage.

A. *Experiment A - Near Field small magnetic loop antenna*

The Near Field setup employed a small magnetic loop antenna positioned 0.5 cm above the central region of the FPGA chip (see Fig. 5). The antenna's position was secured using a stabilizer (PanaVise 381 Vacuum Base with Nylon Jaws) to ensure consistent and stable placement throughout the experiments.

The transmitting antenna was configured to perform a frequency sweep across range of frequencies in order to identify the frequency at which the signal-to-noise ratio (SNR) reached its maximum value. This was our benchmark each antenna's performance.

Fig. 6 presents the signal-to-noise ratio (SNR) measured across frequency for each of the five FPGA antenna architectures implemented in this study: dipole, antenna array, spiral, snake, and fractal configurations. For each architecture depicted in the legend, the antenna routing was synthesized on the FPGA, and electromagnetic emissions were captured in the near field using a small magnetic loop antenna as the receiver. To systematically explore the frequency response, the programmable phase-locked loop (PLL) within the Spartan 6 FPGA was used to generate the carrier, with a maximum output frequency of 936 MHz. The output frequency was then iteratively divided using the a frequency divider, yielding harmonic emissions at integer divisions (e.g., the PLL frequency divided by 2, 3, 4, etc.). This allowed for SNR measurement over a broad frequency spectrum for every antenna architecture. The results highlight not only the distinct frequency-dependent SNR profiles of different antenna layouts, but also the practical emission range achievable using standard near-field detection equipment.

B. *Experiment B - Far Field Dish Antenna*

In the initial far-field experimental setup, a large dish antenna was employed and oriented directly toward the FPGA board. The FPGA board itself was mounted in a vertical position, with its front side directed toward the antenna to maximize signal reception and measurement consistency.

The far-field antenna originally utilized in this setup was designed for operation at 2.4 GHz. However, the maximum achievable frequency for the FPGA, when using it's PLL as the clock source, was limited to 936 MHz. Consequently, this configuration is suboptimal. Despite that, the highest signal-to-noise ratio (SNR) was obtained at 936 MHz in our NF measurements. Therefore, the system was configured to operate at 936 MHz for this experimental setup to ensure optimal performance and measurement accuracy.

In this experiment, the variable parameter was the distance (in meters) between the antenna and the measurement point, while the signal-to-noise ratio (SNR) in decibels (dB) was recorded at the receiving end. The objective was to identify the maximum SNR achievable in the far-field region. To this end, a dipole antenna was employed due to its ability to provide the highest signal-to-noise ratio (SNR) values at frequencies above 200 MHz. This conclusion is supported by the data presented in Fig. 7.

Fig 7 shows the measured signal-to-noise ratio (SNR) as a function of distance for signals transmitted via a routed dipole antenna on the FPGA and received using a TP-Link TL-ANT2424MD dish antenna. As the distance increases, the SNR decreases rapidly at short range and then gradually reaches an asymptote of approximately 20 dB, indicating robust signal detectability even at extended separations.

C. *Experiment C - Far Field Log Periodic Antenna*

In this phase of the far-field experiment, a Log-Periodic Antenna, specifically the HyperLog model 30100X, was positioned to face directly toward the vertically mounted FPGA board. The Log-Periodic Antenna operates over a frequency range of 380 MHz to 10 GHz. For transmission, a dipole antenna was selected due to its superior signal-to-noise ratio (SNR) performance at higher frequencies. Additionally, to maintain consistency and enable meaningful comparisons, the experimental frequency was set to 936 MHz. As illustrated in Fig. 8, the SNR was measured at increasing distances between the FPGA and the receiver, demonstrating a characteristic rapid decline in SNR at shorter ranges, followed by stabilization at longer distances. These results confirm the system's robust signal detectability in far-field scenarios even with increasing distances.

V. Filters

In this section, we focus on the design and implementation of filtering approach aimed at suppressing Gaussian noise, as well as other unwanted noise components captured alongside the desired signal. The primary objective of this filtering process is to maximize the signal-to-noise ratio (SNR) in far-field reception measurements.

As shown in Fig. 9, we first apply a histogram-based Wiener filter to the received spectrum [13]. The noise floor is estimated as the mode of the power histogram, and bins dominated by noise are attenuated while the narrowband signal is preserved. A band-pass filter is then first applied around the target frequency to further enhance the effective signal-to-noise ratio. Subsequently, a specialized filter is used to

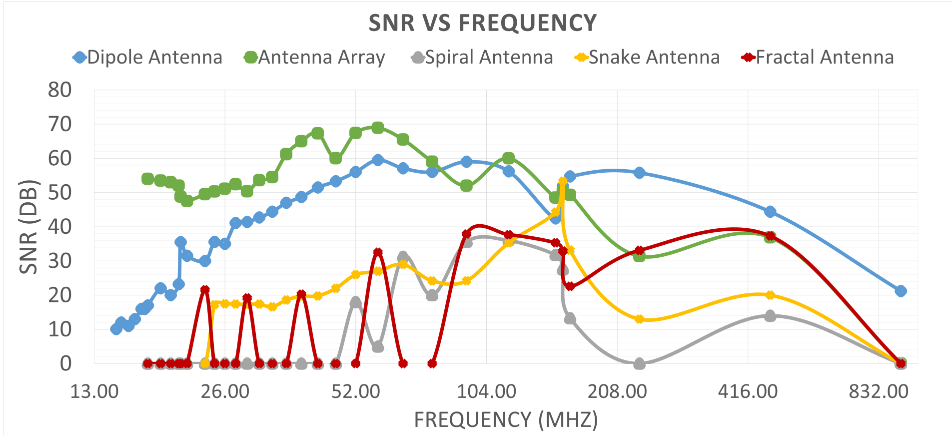

Fig. 6. SNR vs Frequency in Near Field measurements

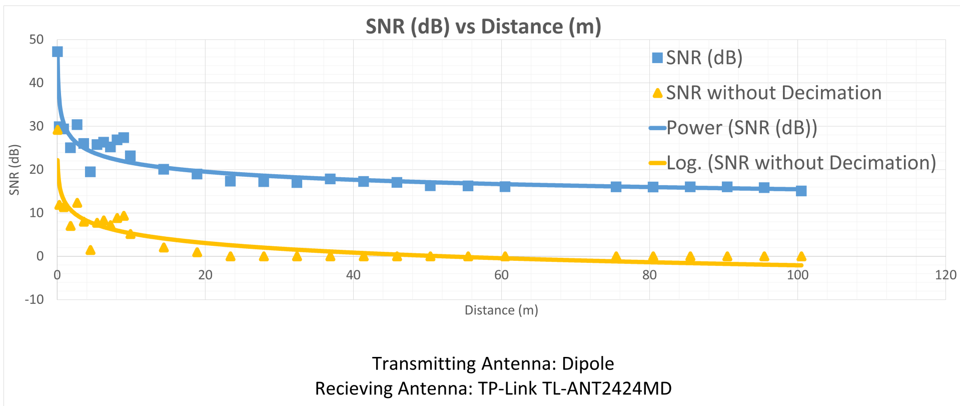

Fig. 7. SNR vs Distance in Dish Antenna

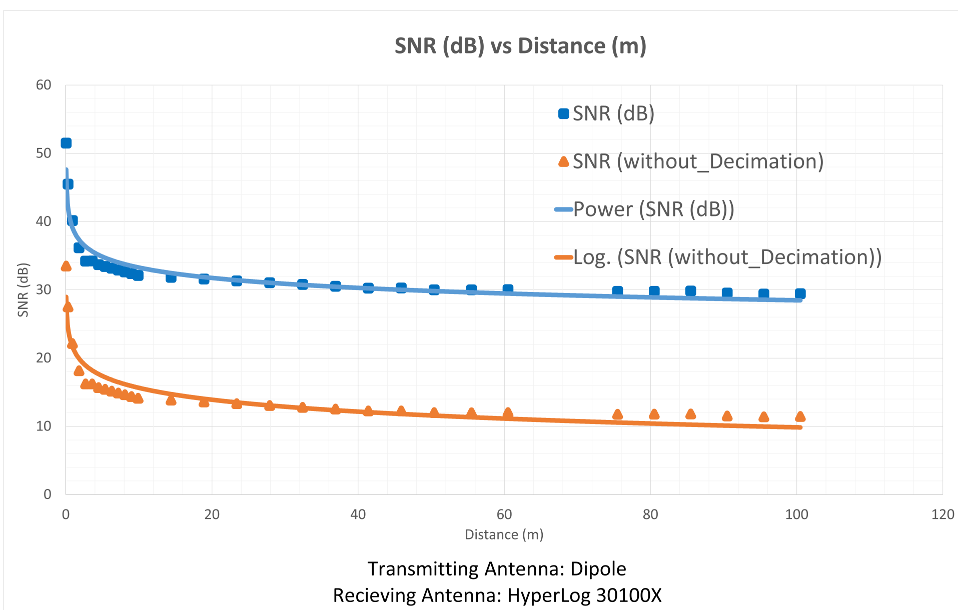

Fig. 8. SNR vs Distance in Log Periodic Antenna

attenuate the known oscillator harmonics, ensuring they are not mistakenly detected as part of the transmitted signal. In this example, a 41dB attenuation was applied to frequencies between 912.499 MHz and 912.501 MHz.

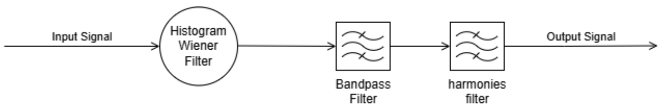

Fig. 9. Simple filter chain design

Figure 10 presents the original spectrum alongside the filtered output for the low voltage (zeros) case, while Figure 11 shows corresponding results for the high voltage (ones) case. These figures illustrate the filter's effectiveness in removing external noise and oscillator harmonies while enhancing the transmitted signal. The transmission was centered at 912 MHz, a frequency range affected by harmonics from the board's 48 MHz oscillator, resulting in higher noise levels. The exact transmitted signal frequency was 912.501 MHz, providing a clear demonstration of the filter's performance.

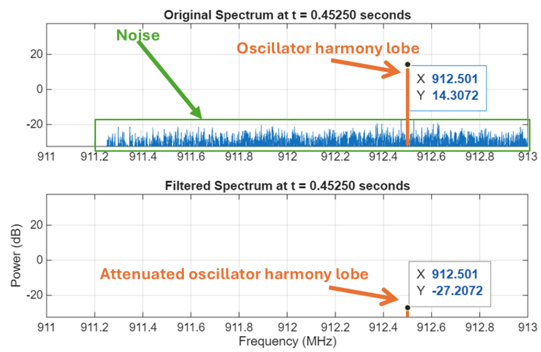

Fig. 10. Transmitted Signal is 0 (low voltage)

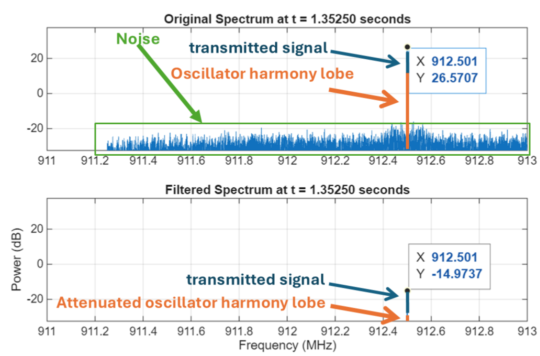

Fig. 11. Transmitted Signal is 1 (high voltage)

VI. Signal Modulation

In order to demonstrate leakage of real data, amplitude shift keying (ASK) and frequency shift keying (FSK) were employed to transmit a 128-bit message (that could be a secret key).

In our updated FPGA architecture in Fig. 12, The top-level module implements Frequency Shift Keying (FSK) by instantiating two separate PLLs (pll_888 and pll_936) that generate carrier frequencies of 888 MHz and 936 MHz respectively from a 48 MHz input clock. A modulator module receives a 128-bit bitstream, symbol timing (bits per symbol duration), and repetition factor via a 19-byte UART packet, then serially outputs this data MSB-first with configurable bit duration and repetition. The modulator's output drives the freq_select signal that switches between the two PLLs (0→888MHz, 1→936MHz), encoding binary data as frequency transitions. A 2-state FSM (IDLE → MODULATE) waits for UART data, triggers the modulator with fsk_start, and sends back an acknowledgment (0xAAAA) when transmission completes (fsk_done). The selected PLL outputs are MUXed together to produce the final FSK-modulated carrier that drives the antenna routing structure, allowing arbitrary data transmission with programmable symbol rates and repetition for error resilience.

The system can be viewed as either Amplitude Shift Keying (ASK) or Frequency Shift Keying (FSK) depending on the receiver's observation perspective. When observing at a single fixed frequency (e.g., monitoring only 888 MHz or only 936 MHz with a narrow-band receiver), the signal appears as ASK - the amplitude toggles between present (carrier on) and absent (carrier off) as the modulator switches frequencies. However, when observing both frequencies simultaneously with a wide-band receiver or spectrum analyzer, the signal appears as true FSK - the carrier frequency shifts between 888 MHz and 936 MHz to encode binary 0s and 1s.

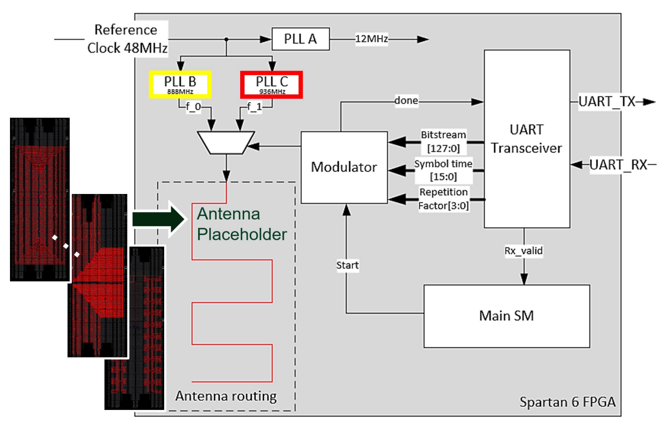

Fig. 12. Block Diagram of the FPGA modulation and control architecture

Following this scheme, Fig. 13 presents the resulting spectrogram of the FSK transmission, where the two intended carrier frequencies-936 MHz for logic '1' and 888 MHz for logic '0'-appear as distinct vertical lines at the far left and far right of the display, respectively. This clear frequency separation visually confirms the successful generation and detection of wide-band FSK modulation using dual PLLs, emphasizing both channel flexibility and the difficulty of intercepting such emissions with conventional narrow-band radio receivers. An additional advantage of this approach is that with more advanced equipment, an adversary could reconfigure the PLLs to select a much larger frequency separation between the FSK tones. This flexibility enables the creation of custom, wideband FSK signals that further reduce the likelihood of detection by conventional radio monitoring equipment, making

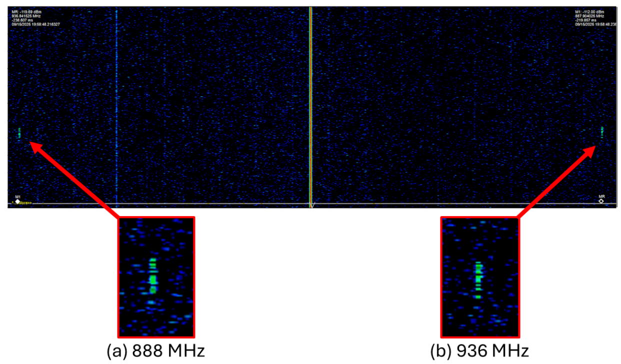

Fig. 13. Spectrogram of FSK, (a) ‘0’ and (b) ‘1’ frequencies can be seen in the far left and far right sides respectively.

the covert bitstream transmission even more resilient and difficult to intercept.

A. Bit Error Rate (BER)

In the first phase of the experiment, the message was transmitted directly using the ASK modulation method. The predefined preamble sequence was “10101010” to facilitate accurate identification of the message start at the receiver. Due to the high transmission speed employed, the use of SDR decimation was not feasible, which resulted in the observed signal-to-noise ratio (SNR) trend corresponding to the orange curve depicted in Fig. 15.

B. Error Correction Codes (ECC)

As shown in Fig. 15, the success rate exhibited a noticeable decline as the transmission distance increased. To address this degradation, a 5-repetition error correction coding (ECC) scheme was implemented. The adoption of this simple ECC method successfully enhanced the reliability of the transmitted data, resulting in a significant improvement in the success rate over extended distances.

Channel Characterization: Comprehensive channel characterization was conducted at a distance of 10.8 meters, employing a dipole antenna as the transmitter and a HyperLOG® 30100X antenna as the receiver. The primary objective of this measurement was to evaluate the error performance of the received signals.

The error transition probabilities were estimated based on the received data vs expected (profile) data as shown in Fig. 14. The results are summarized in the following channel model diagram highlighting a rather symmetric channel and therefore a non symmetric code was not needed and a simple symmetric repetition ECC was sufficient.

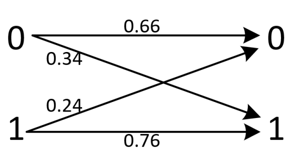

Fig. 14. Channel model

Though these results indicate only slightly asymmetric error behavior between the two transmitted symbol states. In a bit-biased symmetric channel (BSC), the signal-to-noise ratio (SNR) of the higher frequency representing the ‘1’ symbol can vary significantly. Wider-spectrum modulation schemes, which use distinct frequencies with differing SNR levels, provide an opportunity to design more adaptive error correction codes (ECC). By prioritizing symbols experiencing reduced SNR, such an ECC system can enhance overall communication reliability and efficiency.

*C. **Trojan Example:** Rare Triggering attacking a Decryption Oracle*

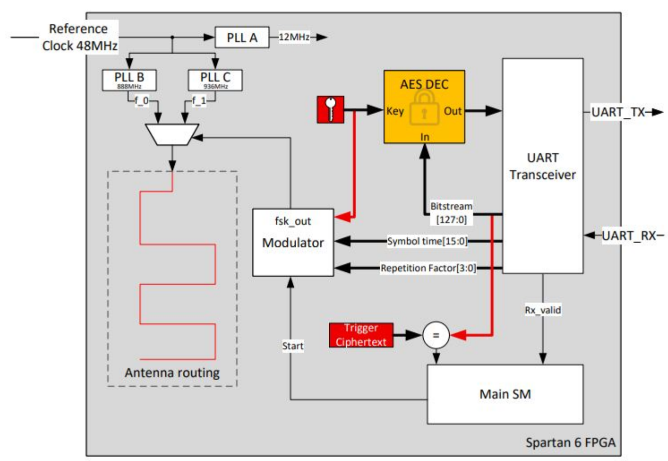

Fig. 16. Block Diagram of the enhanced modulation architecture with added AES block

Figure 16 architecture enhances the design presented in Fig. 12 by introducing a decryption oracle capability, strengthening its relevance for cryptographic security research. Specifically, the integration of the decryption oracle allows the system to accept arbitrary ciphertexts for decryption, enabling the practical demonstration of an adaptive chosen ciphertext attack (CCA) in a hardware setting. Through this enhancement, it becomes possible to simulate a powerful CCA adversary who can send specially crafted “trigger ciphertexts” to the system.

When the rare trigger ciphertext is received, the system’s modulator leaks critical key-related information through our antenna. Because this trigger is activated infrequently and only under specific conditions, detecting or catching such an attack becomes especially challenging. This setup convincingly demonstrates how the addition of a decryption oracle opens a pathway for a CCA adversary to exploit side-channel vulnerabilities in configurable hardware, ultimately facilitating key leakage in a way that would be nearly impossible to observe using regular ciphertexts or standard interfaces.

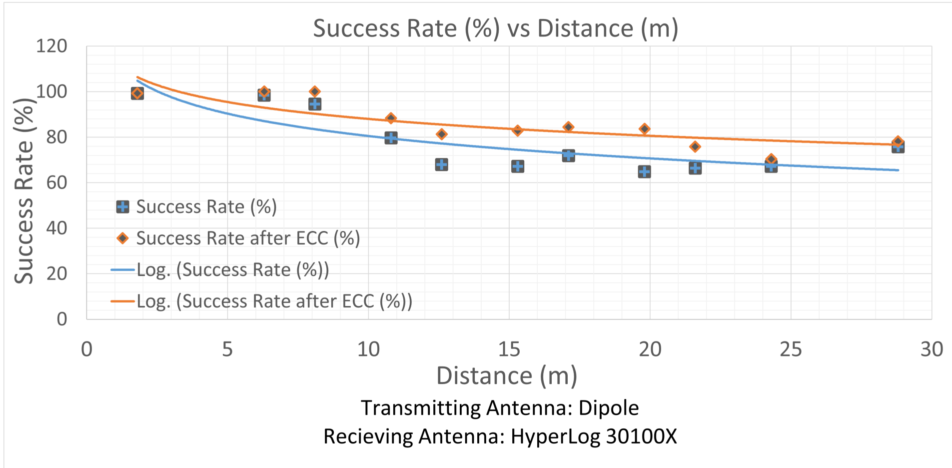

Fig. 15. Bit Error Rate vs Distance

Fig. 17. FPGA-editor view of the AES routing in blue, vs the antenna routing in yellow

Figure 17 illustrates the distinct routing paths used for the AES module and the antenna module within the FPGA fabric. The design demonstrates that both modules can coexist within the same device, making efficient use of available routing resources. Despite their potentially competing placement and interconnect requirements, the AES cryptographic core and the antenna path are able to pass place-and-route processes successfully, confirming that secure logic and RF signal paths can be implemented together in a single architecture without layout conflicts, even with this tiny FPGA chip (with larger chips much larger antennas or far more hidden, relatively, are possible). This validates the feasibility of integrating cryptographic and communication functionalities on the same FPGA device.

VII. Adversary Model

We formalize the adversary in order to ground the threat analysis and to enable assessment of defenses under explicit capability sets.

A. Adversary Classes

- **Local Malicious Tenant (LMT):** A tenant that uploads and executes a bitstream on a shared FPGA (as in cloud FPGA offerings) and can fully control logic, placement hints, and run-time behavior of its design [14] [15].
- **Insider/Developer (ID):** A privileged party with development or maintenance access to on-premise devices; may inject radiator logic in otherwise trusted images [16].
- **Colluding Tenants (CT):** Two or more tenants collude: one transmits embedded radiation, another uses nearby hardware (or cloud-provided adjacent resources) to receive and decode signals [17].
- **Firmware-Assisted Adversary (FAA):** An adversary that combines bitstream modifications with compromised board firmware (bootloaders, JTAG) to relax constraints or gain more precise timing/clock control [18].

Adversary Goals and Metrics: Primary goals include *exfiltration* (secret leakage), *covert command-and-control*, and *side-channel reconnaissance* (probing cryptographic modules). Performance metrics: achievable throughput (bits/s), detectable SNR at distance, stealthiness (probability of detection by static/dynamic screening), and resilience to co-tenancy noise.

Assumptions and Limits: We assume the adversary cannot physically alter package-level shielding or the board's macroscopic RF chain, and cannot access dedicated RF front-ends. We assume the cloud provider restricts bitstream size and run-time duration but may not comprehensively screen for radiative nets. We also assume typical router heuristics and place-and-route nondeterminism may reduce but not eliminate the adversary's ability to obtain dense routed geometries-especially when aided by placement constraints and dont_touch directives.

VIII. Threat Vectors and Attack Scenarios

We enumerate several *plausible* and *novel* attack vectors that build on fabric-only radiators, emphasizing multi-tenant and hybrid attacks.

1) **Single-Tenant Covert Exfiltration:** A malicious tenant embeds a fabric-only antenna in an accelerator image. The radiator transmits small key material or model weights periodically; an external receiver in the vicinity (malicious insider, parked vehicle, or nearby rooftop) collects transmissions. Low duty cycles and repetition codes enable exfiltration at very low average power [19].
2) **Cross-Tenant Collusion:** Two tenants collude: transmitter T1 embeds a radiator, while receiver T2 (on a neighboring board or VM instance but physically nearby) instruments its logic to act as a synchronized receiver or to forward SDR-like samples via side-channels (e.g., network timing) to an external collector. This bypasses cloud-level isolation [20].
3) **Harmonic Hopping and Multiband Evasion:** An adversary cycles carrier energy across multiple harmonics to evade narrowband spectral monitors. Harmonic hopping combined with ultra-low duty cycles makes detection via single-band power monitors ineffective, while harmonic summing at a collocated receiver recovers the signal coherently [21].
4) **Firmware-Enhanced Radiators:** When firmware/bootloaders are compromised (FAA), the adversary can: raise PLL frequencies, disable spectral monitors, adjust board clocking to enhance coherence, or relax placement-time constraints by patching bitstreams post-deployment-leading to more efficient radiators and longer ranges [22].
5) **Insider-Assisted Supply Chain Leakage:** Engineers with bitstream signing capabilities embed radiators in test images that are later promoted to production. These radiators remain dormant until an activation signal (in-band or out-of-band) triggers micro-bursts of emission during non-secure maintenance windows [23].
6) **Far-Field Localization and Tracking via Multi-SDR Adversarial Sensing:** An adversary equipped with three spatially separated SDR receivers can accurately localize a single FPGA emitting fabric-based harmonics, even when the radiated power is extremely low. By combining RSSI measurements, phase offsets, and time-difference-of-arrival analysis, the attacker can perform precise triangulation and track the device in real time. This enables following a person carrying the FPGA, monitoring a vehicle in which the board is installed, or even pinpointing the location of a concealed server room relying on physical obscurity. In effect, the FPGA becomes an unintended RF beacon, providing GPS-like tracking capabilities without any cooperation from the system. Such localization constitutes a powerful reconnaissance vector that bypasses all digital or logical security mechanisms.

IX. Detection and Mitigation Strategies

We envision that future work should be focus on layered defenses that combine static analysis, design-time controls, runtime monitoring, and site-hardening.

Design-Time Controls:

- **Bitstream linting:** Static checks for long constraint lists, suspicious dont_touch usage, and unnatural placement patterns (near-linear long nets spanning many CLBs).
- **Router-aware heuristics:** Integration with place-and-route tools to warn when routing resources are consumed in radiative geometries (e.g., contiguous long nets, high repeater density).
- **Restricted primitives:** Disallowing unchecked PLL instantiation or high-frequency internal clocks in tenant bitstreams unless authorized.

Runtime Monitoring:

- **Spectral monitoring:** Deploy multi-band RF sensors to watch for unusual emissions near FPGA racks. Combine narrowband and broadband monitoring with anomaly detection that accounts for harmonic structure.
- **Power and EM fingerprints:** Build baseline signatures for benign workloads; flag deviations that correlate with pulsed or periodic high-frequency activity.
- **Telemetry:** Instrument FPGA toolchains to log placement constraints and PLL usage; correlate logs with runtime power/spectral telemetry for forensic analysis.

Policy and Infrastructure:

- **Physical controls:** Shielding, RF absorbers around racks, and controlled physical access reduce the attack surface for far-field collection.
- **Proof-of-work in bitstreams:** Use certified, signed bitstreams for critical deployments and require runtime attestation of design provenance.

X. Discussion and Conclusions

In our work, we demonstrated that it is possible to recover information from an FPGA solely through the exploitation of its internal wiring, without the need for any dedicated RF transmission hardware. By leveraging the programmable interconnects to form structured, radiative nets, we successfully detected emitted signals at noticeable-and previously unreported-distances from the device. This approach opens up new possibilities for contactless signal extraction and side-channel analysis on FPGAs [24], [25]. Additionally, by incorporating advanced error correction coding (ECC) techniques, we were able to substantially reduce the bit error rate (BER), further enhancing the reliability and range of information recovery in noisy and low-power scenarios. Our results show that even electrically short, sub-resonant fabric-only nets can radiate coherent energy when driven and buffered appropriately. The achievable throughput is modest (tens to low hundreds of bits per second under conservative ECC), but this is adequate for key exfiltration or Trojan triggering (where we show one concrete example over a decryption oracle). The multi-harmonic footprint complicates detection, while collusion vectors (phased arrays, cross-tenant receivers) amplify practical risk in cloud and on-premise multi-user contexts.

XI. Future Work

Future work in this field could focus on exploring the impact of increasing internal routing density to further enhance electromagnetic emissions from FPGA-based antenna structures. Currently, our approach utilizes LUTs and the wires connecting them to form geometric shapes within the fabric. By intentionally increasing the amount of interconnect-adding more wiring segments between LUTs or designing denser net structures-it will be possible to investigate the correlation between increased wiring, signal strength, and radiated power. Systematically studying how the addition of extra routing resources influences transmission characteristics could lead to new design strategies for stronger or more efficient FPGA-based transmitters, as well as deeper understanding of the physical layer properties achievable using modern reconfigurable logic fabrics.

Future research could also focus on developing and employing more advanced receiver antennas and signal processing techniques to improve detection of low-SNR emissions from FPGA-based transmitters. Rather than relying solely on traditional time-domain or single-frequency detection, one promising strategy involves summing multiple harmonics of the transmitted signal-an approach known as harmonic summing-to coherently enhance weak periodic signals that may be buried in noise. Other effective low-SNR techniques include digital matched filtering, autocorrelation, beamforming, and pulse integration, all of which can be implemented using modern digital receivers or FPGA-based platforms. By integrating methods such as greedy or optimized harmonic summing algorithms, adaptive filtering, or synchronized multi-antenna arrays, future systems could achieve significantly higher detection sensitivity and greater resilience to noise or interference, allowing for more robust communication channels-even when radiated signals are extremely weak or dispersed across wide frequency bands. Such efforts would reciprocate more effort in research on prevention and detection.

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